

```
data(iris)
head(iris)
```

```
## Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1          3.5          1.4          0.2 setosa
## 2          4.9          3.0          1.4          0.2 setosa
## 3          4.7          3.2          1.3          0.2 setosa
## 4          4.6          3.1          1.5          0.2 setosa
## 5          5.0          3.6          1.4          0.2 setosa
## 6          5.4          3.9          1.7          0.4 setosa
```

Testing

```
summary(
  lm(Petal.Width ~ Species + Species:Petal.Length, data = iris)
)
```

```
##
## Call:
## lm(formula = Petal.Width ~ Species + Species:Petal.Length, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6337 -0.0744 -0.0134  0.0866  0.4503
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.04822    0.21472   -0.225  0.822627
## Speciesversicolor    -0.03607    0.31538   -0.114  0.909109
## Speciesvirginica     1.18425    0.33417    3.544  0.000532 ***
## Speciessetosa:Petal.Length    0.20125    0.14586    1.380  0.169813
## Speciesversicolor:Petal.Length  0.33105    0.05391    6.141  7.56e-09 ***
## Speciesvirginica:Petal.Length  0.16030    0.04590    3.492  0.000636 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1773 on 144 degrees of freedom
## Multiple R-squared:  0.9477, Adjusted R-squared:  0.9459
## F-statistic: 521.9 on 5 and 144 DF, p-value: < 2.2e-16
```

```
summary(
  lm(Petal.Width ~ Species * Petal.Length, data = iris)
)
```

```
##
## Call:
## lm(formula = Petal.Width ~ Species * Petal.Length, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6337 -0.0744 -0.0134  0.0866  0.4503
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.04822    0.21472   -0.225  0.822627
```

```
## Speciesversicolor          -0.03607    0.31538   -0.114 0.909109
## Speciesvirginica           1.18425    0.33417    3.544 0.000532 ***
## Petal.Length               0.20125    0.14586    1.380 0.169813
## Speciesversicolor:Petal.Length 0.12981    0.15550    0.835 0.405230
## Speciesvirginica:Petal.Length -0.04095    0.15291   -0.268 0.789244
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1773 on 144 degrees of freedom
## Multiple R-squared:  0.9477, Adjusted R-squared:  0.9459
## F-statistic: 521.9 on 5 and 144 DF,  p-value: < 2.2e-16
```

Way 1: Categorical * Numerical

```
way1 <- lm(Petal.Width ~ Species * Petal.Length, data = iris)
summary(way1)
```

```
##
## Call:
## lm(formula = Petal.Width ~ Species * Petal.Length, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6337 -0.0744 -0.0134  0.0866  0.4503
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.04822    0.21472   -0.225 0.822627
## Speciesversicolor -0.03607    0.31538   -0.114 0.909109
## Speciesvirginica  1.18425    0.33417    3.544 0.000532 ***
## Petal.Length    0.20125    0.14586    1.380 0.169813
## Speciesversicolor:Petal.Length 0.12981    0.15550    0.835 0.405230
## Speciesvirginica:Petal.Length -0.04095    0.15291   -0.268 0.789244
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1773 on 144 degrees of freedom
## Multiple R-squared:  0.9477, Adjusted R-squared:  0.9459
## F-statistic: 521.9 on 5 and 144 DF,  p-value: < 2.2e-16
```

Notes: - Gives slope for **baseline** group. - Adjustment terms to *slope* and the *intercept* for the other groups.

Way 2: Categorical + Numerical + Categorical:Numerical

```
way2 <- lm(Petal.Width ~ Species + Petal.Length + Species:Petal.Length, data = iris)
summary(way2)
```

```
##
## Call:
## lm(formula = Petal.Width ~ Species + Petal.Length + Species:Petal.Length,
##     data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6337 -0.0744 -0.0134  0.0866  0.4503
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.04822    0.21472   -0.225 0.822627
## Speciesversicolor    -0.03607    0.31538   -0.114 0.909109
## Speciesvirginica      1.18425    0.33417    3.544 0.000532 ***
## Petal.Length        0.20125    0.14586    1.380 0.169813
## Speciesversicolor:Petal.Length  0.12981    0.15550    0.835 0.405230
## Speciesvirginica:Petal.Length -0.04095    0.15291   -0.268 0.789244
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1773 on 144 degrees of freedom
## Multiple R-squared:  0.9477, Adjusted R-squared:  0.9459
## F-statistic: 521.9 on 5 and 144 DF,  p-value: < 2.2e-16
```

Notes: - Exactly the same as way 1

Way 3: Categorical + Categorical:Numerical

```
way3 <- lm(Petal.Width ~ Species + Species:Petal.Length, data = iris)
summary(way3)
```

```
##
## Call:
## lm(formula = Petal.Width ~ Species + Species:Petal.Length, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6337 -0.0744 -0.0134  0.0866  0.4503
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.04822    0.21472   -0.225 0.822627
## Speciesversicolor    -0.03607    0.31538   -0.114 0.909109
## Speciesvirginica      1.18425    0.33417    3.544 0.000532 ***
## Speciessetosa:Petal.Length  0.20125    0.14586    1.380 0.169813
## Speciesversicolor:Petal.Length  0.33105    0.05391    6.141 7.56e-09 ***
## Speciesvirginica:Petal.Length  0.16030    0.04590    3.492 0.000636 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1773 on 144 degrees of freedom
## Multiple R-squared:  0.9477, Adjusted R-squared:  0.9459
## F-statistic: 521.9 on 5 and 144 DF,  p-value: < 2.2e-16
```

Notes: - Separate slope terms for each category. - Adjustment terms to intercept but *not* to slope. - No initial slope, just absolute slopes

No Interaction Terms: Categorical + Numerical

- Different lines from previous models
- Different intercepts, but same slope for all 3 lines
- Not the same equations as above

```

way.noint <- lm(Petal.Width ~ Species, data = iris)
summary(way.noint)

##
## Call:
## lm(formula = Petal.Width ~ Species, data = iris)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.626 -0.126 -0.026  0.154  0.474
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.24600    0.02894     8.50 1.96e-14 ***
## Speciesversicolor  1.08000    0.04093    26.39 < 2e-16 ***
## Speciesvirginica   1.78000    0.04093    43.49 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2047 on 147 degrees of freedom
## Multiple R-squared:  0.9289, Adjusted R-squared:  0.9279
## F-statistic: 960 on 2 and 147 DF, p-value: < 2.2e-16

```

Model Selection

```

finches <- read.csv("https://raw.githubusercontent.com/njlytal/MATH314/main/finches.csv")
str(finches)

```

```

## 'data.frame':   68 obs. of  9 variables:
## $ island      : chr  "santacruz" "santacruz" "santacruz" "santacruz" ...
## $ winglength  : num  63.5 69.5 70 70 70.5 66 68 68.5 68.5 72 ...
## $ taillength  : num  40 41 44 43.5 42 41 40 43.5 40 44.5 ...
## $ beakwidth   : num  9 9 9.2 9.2 9.5 10 8.5 10.2 10 10.8 ...
## $ beakheight  : num  11 12 12 11.2 12 12.2 12 13 12.5 13 ...
## $ lowerbeaklength: num  9 8.5 8.5 8.5 8.5 9.2 8.5 9.2 9 9.2 ...
## $ upperbeaklength: num  16 16 16 16.2 16.5 17 17 17 17.5 17.5 ...
## $ tarsuslength : num  20 21.2 21 20.5 22 22 22 22 22 21.2 ...
## $ middletoelength: num  18.5 19.5 18.6 19 19.2 18 17.8 19.2 20 19 ...

```

Goal: Create a model to predict winglength based on all numerical vars

```

model <- lm(
  winglength ~ taillength +
    beakwidth + beakwidth +
    beakheight + lowerbeaklength +
    upperbeaklength + tarsuslength +
    middletoelength,
  data = finches
)
summary(model)

```

```

##
## Call:
## lm(formula = winglength ~ taillength + beakwidth + beakwidth +
##      beakheight + lowerbeaklength + upperbeaklength + tarsuslength +

```

```

##      middletoelength, data = finches)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -3.8597 -1.4918  0.0955   1.3891   3.8927
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    30.1162     8.5839   3.508 0.000861 ***
## taillength      0.3756     0.1481   2.536 0.013829 *
## beakwidth       0.6293     0.5658   1.112 0.270424
## beakheight      0.8114     0.4873   1.665 0.101091
## lowerbeaklength -0.2296     0.8271  -0.278 0.782288
## upperbeaklength -0.1383     0.4395  -0.315 0.754094
## tarsuslength    0.1420     0.3545   0.400 0.690245
## middletoelength  0.4903     0.3583   1.368 0.176264
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.951 on 60 degrees of freedom
## Multiple R-squared:  0.7905, Adjusted R-squared:  0.7661
## F-statistic: 32.35 on 7 and 60 DF,  p-value: < 2.2e-16

```

```

model <- lm(
  winglength ~ taillength +
    beakheight,
  data = finches
)
summary(model)

```

```

##
## Call:
## lm(formula = winglength ~ taillength + beakheight, data = finches)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -4.1864 -1.1769  0.0875   1.3021   4.1441
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    37.5372     4.5784   8.199 1.30e-11 ***
## taillength      0.4145     0.1341   3.092 0.00293 **
## beakheight      1.2269     0.1618   7.582 1.61e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.932 on 65 degrees of freedom
## Multiple R-squared:  0.7776, Adjusted R-squared:  0.7707
## F-statistic: 113.6 on 2 and 65 DF,  p-value: < 2.2e-16

```